Square Elliptic Completions and APS Boundary Information on Round Spherical Quotients

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Abstract

The Atiyah–Patodi–Singer correction combines spectral asymmetry with zero modes. We distinguish a sparse elliptic completion (one permitted off-diagonal block is set to zero) from its fully coupled twisted-Dirac completion, and derive the latter's signed round-boundary spectrum from its homogeneous connection. For its round-lens character transform we derive a closed Dedekind-sum phase law at every rank and modulus, prove that rank one generates the entire exponentiated ladder, and give a necessary-and-sufficient criterion for equality under a coherent character relabeling. We also prove infinite families in which two adjacent ranks lose all dependence on the lens parameter, including composite moduli. The 23,688 exact table entries remain reproducible by two routes. The sparse and Dirac completions are elliptically homotopic, but no equality of their boundary spectra is asserted. We supply neither a numerical index on a chosen filling nor a physical interpretation of the phase.

Scope. This paper supplies an operator-level derivation, not an identification inferred from a character match. Sections 2–4 specify the full completion, its connection, chirality, collar and signed spectrum; the subsequent phase proofs use the resulting transform. No chosen-filling index, off-round extension or physical phase is claimed.

1 Introduction

Elliptic boundary problems naturally separate an interior symbol question from a boundary-information question. The first asks whether an overdetermined differential operator can be completed to a square elliptic one. The second asks how much geometry survives in its spectral boundary correction. For the operators considered here these questions meet in a particularly explicit form: adding the missing cross-block to a sparse Stein–Weiss completion and normalizing the columns gives a twisted-Dirac operator, whose round boundary operator contains the tensor-curl family as one reducing summand. The remaining summand is not spectrally negligible; its eta and kernel contributions are essential to the APS correction.

Let $(a, b)_{\mathrm{str}}$ denote the irreducible complex structure-group $\mathrm{Spin}(4)$ representation $V_a \boxtimes V_b$, where V_m is the $(m+1)$-dimensional irreducible $SU(2)$ representation. The integer $J \geq 1$ is the tensor

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rank in the normalization used below. Unsubscripted pairs in Section 2 are structure labels; pairs describing boundary eigenspaces from Section 3 onwards are isometry labels. Their relation is stated in Section 3.1. Our objects are a sparse square elliptic operator $\mathcal{S}_J$, a full completion $\mathcal{G}_J$, its Clifford normalization $D^+_{T_J}$, the latter's intrinsic self-adjoint boundary operator B_J, and the tensor-curl block A_J on a free round quotient $Y = S^3/\Gamma$. Flat twists are denoted by ρ.

The argument has three stages. Sections 2–3 identify the full elliptic completion and derive the first-order round operator and reducing split. Sections 4–5 assemble the complementary eta and kernel, derive a cumulative formula with the companion tensor-curl and McKay identities, and evaluate a finite lens-space reference table by two independent exact routes. Sections 6–7 use the resulting class transforms to compare real eta data, reduced APS corrections, and their exponentials.

Generalized-gradient ellipticity and higher-spin Dirac factorization on constant-curvature spaces are established machinery [Pil11, arXiv v1, p. 1 and Thms. 3.12–3.13] [HT21, Prop. 1 and Thms. 5, 7, 10]. Explicit spin-Dirac spectra and twisted eta formulas on round spherical space forms are likewise classical [Bär96, Thms. 1–2] [BL17, Thm. 4.3] [Gil84a, Lemmas 2.1–2.2 and Thm. 2.5] [CM01, ĆMT14]. We use those frameworks at the point where the exact normalization-specific calculation enters.

The principal statements are as follows.

(i) The specified overdetermined Stein–Weiss operator has an explicit square elliptic completion for every integer $J \geq 1$, with a closed positive determinant-square law.

(ii) On free round spherical quotients,

$$B_J \cong A_J \oplus B_J^{\mathrm{aux}}$$

is an orthogonal reducing split. The eta invariant and kernel dimension of B_J^{aux} have exact all-rank representation-ring formulas.

(iii) An exact calculation covers every $2 \leq p \leq 24$, every unit q, both spin lifts where distinct, $J = 1, \ldots, 6$, and every flat character. It contains 23,688 exact triples (η, h, ξ), with independent rank-seven and rank-eight checks.

(iv) The complete class transforms of η, h, and $\xi = (\eta + h)/2$ are rational character expressions. Reduced ranks one and two have universal nonidentity-class support, while the rank-one reconstruction theorem itself is known.

(v) Exact finite computation refutes universal oriented adjacent-rank reconstruction. Sixteen rows represent four colliding oriented-orbit pairs; two collisions remain nonhomeomorphic even after orientation reversal.

(vi) For the derived transform, a Dedekind-sum formula gives every phase, rank one generates every higher phase, and a complete coherent-label collapse criterion follows. The pair $L(16, 1)$ and $L(16, 9)$ is a corollary. Infinite blind adjacent-rank families include admissible composite moduli.

The arithmetic uses classical Fourier–Dedekind sums and congruences; no priority claim follows from the local verification. The transform-specific deductions are separated from those inputs and from any filling-index claim.

2 The square completion

2.1 A sparse elliptic square

The generalized-gradient construction begins with the Levi-Civita covariant derivative and orthogonal projection onto irreducible Spin(4) summands. Pilca's ellipticity classification provides the ambient framework [Pil11, arXiv v1, p. 9, Thms. 3.12–3.13]. The essential data here are the exact representation choice, column normalization, and determinant law.

For $J \geq 2$, choose the right-handed mirror of the sparse completion and set

$$\mathcal{S}_J : (J, J) \oplus (J, J-2) \longrightarrow (J+1, J-1) \oplus (J-1, J-1).$$

Write $E = (J, J)$, $C = (J, J-2)$, $F_1 = (J+1, J-1)$ and $F_2 = (J-1, J-1)$. If $P_{UV}\nabla$ denotes covariant differentiation followed by the orthogonal projection from U to V, the sparse operator is

$$\mathcal{S}_J = \begin{pmatrix} P_{EF_1}\nabla & 0 \\ P_{EF_2}\nabla & P_{CF_2}\nabla \end{pmatrix}.$$

There are three nonzero entries, not four. In a left-handed presentation the auxiliary summand is $(J-2, J)$; the two choices are mirrors, not distinct ellipticity statements. The source and target both have complex rank $2J(J+1)$. At $J = 1$ the negative-label auxiliary summand is absent and the same statement is read in the reduced two-summand convention.

Fix a unit covector. Restriction to its diagonal $SU(2)$ stabilizer makes the symbol block triangular. With orthonormal irreducible bases, the diagonal amplitude squares are

$$\alpha_r^2 = \frac{r(r+1)}{2(J+1)^2}, \qquad 1 \leq r \leq J, \tag{1}$$

$$\beta_r^2 = \frac{(J-r)(J+r+1)}{2(J+1)^2}, \qquad 0 \leq r \leq J-1, \tag{2}$$

$$\gamma_r^2 = \frac{r(r+1)}{2(J^2-1)}, \qquad 1 \leq r \leq J-1. \tag{3}$$

Every factor in its actual range is positive. Stabilizer transitivity and symbol homogeneity then extend invertibility from the chosen unit covector to every nonzero covector.

Theorem 2.1 (Exact square completion)**.** *For every integer $J \geq 1$, the operator $\mathcal{S}_J$ is square elliptic. For $J \geq 2$, its invariant determinant-square is*

$$\left(\frac{J}{2(J+1)}\right)^{2J+2} \prod_{r=1}^{J-1} \left(\frac{r^2(r+1)^2}{4(J+1)^2(J^2-1)}\right)^{2r+1} > 0.$$

Proof. At a unit covector, restriction to the diagonal $SU(2)$ stabilizer makes the symbol block triangular. Its one-copy endpoints are β_0 and α_J; each intermediate two-copy determinant is $\alpha_r\gamma_r$. All are nonzero. These amplitude formulas can be obtained from the spin-half projectors: for spin j, the projections in $\frac{1}{2} \otimes j$ are $(2\boldsymbol{S}\cdot\boldsymbol{J} + j + 1)/(2j+1)$ and $(j - 2\boldsymbol{S}\cdot\boldsymbol{J})/(2j+1)$. At a unit covector the vector representation is the normalized singlet of its two spin halves. Thus $\langle 2\boldsymbol{S}_L \cdot \boldsymbol{J}_L\rangle = \langle 2\boldsymbol{S}_R \cdot \boldsymbol{J}_R\rangle = 0$ and their product has expectation $-\boldsymbol{J}_L \cdot \boldsymbol{J}_R$. Substitute $2\boldsymbol{J}_L \cdot \boldsymbol{J}_R = r(r+1) - j_L(j_L+1) - j_R(j_R+1)$ into the two projectors to obtain the displayed squares. The symbol

is therefore invertible at that covector. Stabilizer transitivity and homogeneity give invertibility at every nonzero covector, proving square ellipticity. Multiplying the squared diagonal factors with their stabilizer multiplicities $2r+1$ gives the stated determinant-square formula. For $J=1$ the absent auxiliary column is omitted and the same argument applies to the reduced block. □

This formula belongs to the stated sparse normalization. It is not detached from the operator and advertised as a general determinant construction.

2.2 The full completion and its twisted-Dirac identification

Define the half-spinors by

$$S^+ = (0,1)_{\mathrm{str}}, \qquad S^- = (1,0)_{\mathrm{str}}, \qquad \mathrm{Sym}^2 S^\pm = \Lambda^2_\pm.$$

These definitions incorporate the structure/isometry dictionary of Section 3.1; the superscripts are not the order of the structure factors. Let $T_J = \mathrm{Sym}^J S^- \otimes \mathrm{Sym}^{J-1} S^+ = (J, J-1)_{\mathrm{str}}$. The chiral twisted-Dirac operator

$$D^+_{T_J} : S^+ \otimes T_J \longrightarrow S^- \otimes T_J$$

has source and target decomposition

$$(J,J) \oplus (J,J-2) \longrightarrow (J+1,J-1) \oplus (J-1,J-1),$$

and uses all four projected gradients. Choose the constant unitary identifications of the irreducible summands and their phases so that

$$\mathcal{G}_J = \begin{pmatrix} P_{EF_1}\nabla & P_{CF_1}\nabla \\ P_{EF_2}\nabla & P_{CF_2}\nabla \end{pmatrix}.$$

The omitted entry of $\mathcal{S}_J$ is essential. Multiplying the two source columns of $\mathcal{G}_J$ by

$$c_E = \sqrt{\frac{2(J+1)}{J}}, \qquad c_C = \sqrt{\frac{2(J-1)}{J}},$$

with the second column absent at $J=1$, gives exactly the Clifford symbol. In intermediate diagonal spin $1 \le r < J$, the recoupling matrix is

$$R_{J,r} = \begin{pmatrix} \sqrt{t_r} & -\sqrt{1-t_r} \\ \sqrt{1-t_r} & \sqrt{t_r} \end{pmatrix}, \qquad t_r = \frac{r(r+1)}{J(J+1)}.$$

The column norms follow from the projector formulas above; orthogonality then fixes the second column as $(-\sqrt{1-t_r}, \sqrt{t_r})^T$. At spins $0, J$ the normalized one-copy symbol is an isometry. Equivariance gives the Clifford symbol at every covector. Moreover, this is an equality of operators, not just symbols:

$$D^+_{T_J} = \mathcal{G}_J \,\mathrm{diag}(c_E, c_C).$$

Both sides are the same parallel bundle map composed with the same Levi-Civita-induced connection. There is no adjustable zeroth-order term. The four-block framework is classical [HT21, Sec. 2, Eq. (2.3)]; the normalization is fixed here.

For clarity, the determinant-square of the unnormalized full completion is

$$|\det \sigma(\mathcal{G}_J)|^2 = \left(\frac{J}{2(J+1)}\right)^{(J+1)^2} \left(\frac{J}{2(J-1)}\right)^{J^2-1}, \qquad J \geq 2,$$

whereas the normalized Clifford determinant-square is 1 at a unit covector. These are not the sparse determinant. At $J = 2$ the sparse and full values are respectively $1/14348907$ and $1/19683$.

There is nevertheless an elliptic homotopy: in a normalized intermediate block multiply the upper-right entry of $R_{J,r}$ by $s \in [0, 1]$. Its determinant is $t_r + s(1 - t_r) > 0$, and the endpoints are unchanged. This preserves the interior symbol class. It does not identify the sparse operator's boundary spectrum with that of $D^+_{T_J}$. From now on B_J always refers to the fully coupled Dirac completion.

Although $S^{\pm}$ and T_J individually require spin lifts, their tensor products above descend to $SO(4)$: the diagonal central element $(-1, -1) \in \mathrm{Spin}(4)$ acts by $(-1)(-1) = 1$. The operator therefore exists on an oriented Riemannian four-manifold without a chosen global spin structure, using these associated bundles. Spin notation gives its local factorization. On the boundary a change of spin lift acts on both factors and cancels, explaining the spin-lift independence of the tensorial spectrum.

3 The round boundary operator

3.1 Structure labels versus isometry labels

On $\mathbb{H}$ oriented by $(1, i, j, k)$ use the isometry action $q \mapsto zqw^{-1}$, with the left factor first. Left infinitesimal multiplications correspond to the self-dual two-forms $e^{01}+e^{23}$ and their cyclic partners; right multiplications correspond to the anti-self-dual forms $e^{01} - e^{23}$ and their partners. For the d^- choice below, Section 2's target $(2, 0)_{\mathrm{str}}$ is Λ^2_-, so the explicit dictionary is

$$(a, b)_{\mathrm{str}} = (b, a)_{\mathrm{iso}}.$$

These distinguish structure-group labels from isometry labels. For fundamental isometry characters $x = \chi_1(z)$, $y = \chi_1(w)$ and $P = x - y$, the source-minus-target character of the printed symbol is

$$-P_{\mathrm{str}} \chi_J^{\mathrm{str,L}} \chi_{J-1}^{\mathrm{str,R}} = P\chi_{J-1}(z)\chi_J(w).$$

The identity follows directly from $\chi_{r+1} + \chi_{r-1} = \chi_1\chi_r$; it is valid at every rank. At $J = 1$, the independent exterior-algebra calculation reads

$$\frac{\operatorname{tr}\Lambda^1 - \operatorname{tr}\Lambda^2_- - \operatorname{tr}\Lambda^0}{\det(1-g)} = \frac{xy - (y^2-1) - 1}{(x-y)^2} = \frac{y}{x-y}.$$

Thus the swapped symbol term, rather than the same-labelled one, agrees with the proposed fixed-point numerator. Together with Christoffel–Darboux below, its difference from the reduced transform is minus the kernel character. This is a character-level consistency check; a full filling statement is intentionally not inferred from it.

3.2 Collar, Clifford convention, and a first-order calibration

Give Y its boundary orientation and write the product orientation as $dx \wedge \mathrm{vol}_Y$, with x increasing outwards. The inward collar coordinate is $u = -x$. We define B_J by

$$D^+_{T_J} = c(du)(\partial_u + B_J)$$

using the product Levi-Civita connection and the induced connection on T_J. The APS condition annihilates the nonnegative spectral part of B_J; its index correction is $-\xi(B_J)$, using the inward-normal convention of [APS75a, Theorem 3.10, p. 57; equations (4.1)–(4.3), pp. 60–62]. A change of the normal factorization without changing this spectral convention is not innocuous.

Here is a calibration that fixes the sign independently of a squared operator. At $J = 1$ the Clifford-normalized completion is $(\sqrt{2}d^-, \delta)$ on one-forms. Write a one-form as $f\,dx + \beta$ and identify an anti-self-dual two-form with $(dx \wedge v - *_Y v)/\sqrt{2}$. In the resulting target coordinates its value is

$$(\partial_x \beta - d_Y f - *_Y d_Y \beta,\ -\partial_x f + \delta_Y \beta).$$

The normal symbol sends (f, β) to $(\beta, -f)$. Factoring with the inward coordinate therefore gives

$$B_1 = \begin{pmatrix} 0 & \delta_Y \\ d_Y & *_Y d_Y \end{pmatrix}. \tag{4}$$

In particular, the co-closed one-form block is positive curl. On the unit quaternions with the boundary orientation, the right-invariant form $dq\,q^{-1}$ has curl $+2$ and type $(2, 0)$; its left-invariant counterpart has curl -2 and type $(0, 2)$. Thus the positive TT branch is $(n+2r, n)$, with eigenvalue $n + r + 1$. This is the convention of the companion tensor-curl papers [Mah26c, Mah26b], not its negative.

For use at every rank, view the unit sphere as the symmetric space $\mathrm{Spin}(4)/\mathrm{Spin}(3)$. The canonical and Levi-Civita connections agree here; the homogeneous Dirac formula has no cubic bracket term [Agr02, Sections 2–3]. The following calculation also fixes the twisting connection, not merely a Casimir eigenvalue. Let $e_k \in \{i, j, k\}$ and take horizontal generators $X_k = (e_k/2, -e_k/2)$; the curve $\exp(tX_k) \cdot 1 = \exp(te_k)$ has unit speed. A section of the associated spinor-twist bundle is an equivariant function on $\mathrm{Spin}(4)$, differentiated horizontally by the canonical connection. In its Peter–Weyl decomposition write L_k, R_k for the Hermitian generators on the contragredient representation of type (p, q). The derivative along X_k is $i(L_k - R_k)$. On the boundary spinor factor put $c_Y(e_k) = 2iS_k$, $S = 1/2$, with the Clifford sign selected by (4). Consequently, on the invariant multiplicity space for a twisting summand V_{2r-1}, the actual first-order operator is

$$D = 2\boldsymbol{S} \cdot (\boldsymbol{R} - \boldsymbol{L}), \qquad S = \tfrac{1}{2}, \quad L = \tfrac{p}{2}, \quad R = \tfrac{q}{2}. \tag{5}$$

Here the diagonal invariance condition says that $\boldsymbol{S} + \boldsymbol{L} + \boldsymbol{R}$ couples to twisting spin $r - 1/2$. The $SU(2)$ duals have the same highest weights, so p, q remain the isometry labels. The horizontal bracket has no horizontal component, hence neither the spin nor twist connection contributes an additional horizontal algebraic term. This derives (5) from the specified connection. The first-order $J = 1$ calibration fixes the remaining overall sign; it is not inferred from the Casimir square used below.

3.3 A reducing tensor-curl sector

On a product collar, the tangential operator of $D^+_{T_J}$ is the intrinsic Dirac operator on

$$S_Y \otimes T_J|_{\mathrm{Spin}(3)} = W_0 \oplus 2W_1 \oplus \cdots \oplus 2W_{J-1} \oplus W_J.$$

The rank-J transverse-traceless compression is the tensor-curl operator A_J. On the round three-sphere its two helicities have $\mathrm{Spin}(4)$ support $(n + 2J, n)$ and $(n, n + 2J)$. A lower-spin bundle W_r, $r < J$, only contains types with $|p - q| \le 2r < 2J$. Equivariance therefore prohibits mixing between the TT helicities and the auxiliary summands.

Theorem 3.1 (Round reducing split)**.** *On S^3 and every free round spherical quotient,*

$$B_J \cong A_J \oplus B_J^{\mathrm{aux}}$$

as an orthogonal reducing split. Consequently, for every flat twist ρ,

$$\eta_\rho(B_J) = \eta_\rho(A_J) + \eta_\rho(B_J^{\mathrm{aux}}), \tag{6}$$
$$h_\rho(B_J) = h_\rho(A_J) + h_\rho(B_J^{\mathrm{aux}}). \tag{7}$$

Proof. The two helicity branches of A_J have Spin(4) types $(n+2J, n)$ and $(n, n+2J)$. Every lower-spin summand W_r, $r < J$, has $|p-q| \le 2r < 2J$, so no irreducible type occurring in the tensor-curl sector occurs in the auxiliary sector. Since B_J is equivariant and self-adjoint, the orthogonal projections onto these inequivalent supports commute with it. This gives the reducing split on S^3; equivariance makes the decomposition descend to a free quotient and persist after a flat twist. Additivity of eta and kernel over an orthogonal direct sum gives the two displayed identities. □

The word "round" is essential. No corresponding reducing theorem is used for Berger or arbitrary boundary metrics.

3.4 Local index density

Let x_+, x_- be the formal half-spinor roots, with $x_+^2 = (p_1 + 2e)/4$ and $x_-^2 = (p_1 - 2e)/4$. The weight-square sum in $\mathrm{Sym}^n \mathbb{C}^2$ is

$$\sum_{k=0}^{n} (n-2k)^2 = \frac{n(n+1)(n+2)}{3}.$$

Expanding $\mathrm{ch}(\mathrm{Sym}^J S^-)\mathrm{ch}(\mathrm{Sym}^{J-1} S^+)$ through degree four and multiplying by $\widehat{A} = 1 - p_1/24 + \cdots$ gives, for the displayed positive-chiral source,

$$[\widehat{A}\,\mathrm{ch}(T_J)]_4 = \frac{J(J+1)}{12}\big((J^2+J-1)p_1 - (2J+1)e\big).$$

On a nonspin manifold the left side denotes the invariant polynomial computed in local spin lifts; the right side is globally defined. On a closed four-manifold this integrates to

$$\frac{J(J+1)}{12}\big(3(J^2+J-1)\tau - (2J+1)\chi\big).$$

At $J = 1$ the expression is $(\tau - \chi)/2$, matching the calibrated $d^- + \delta$ control. We do not choose a filling in this paper, so the local density is not converted into a numerical APS index.

4 Eta, kernel, and the all-rank auxiliary assembly

4.1 Classical quotient machinery and APS normalization

The ordinary spin-Dirac spectrum on S^3 and free spherical quotients is described by character generating functions [Bär96, pp. 74–76, Thms. 1–2]. Lens-space multiplicities admit an explicit

affine-congruence-lattice formula [BL17, arXiv v1, p. 14, Thm. 4.3]. Twisted eta and defect formulas on odd-dimensional spherical space forms, including spin-structure typing, are classical [Gil84a, pp. 427–432]. For twisted Dirac spectra on S^3/Γ we cite Cisneros-Molina's research article [CM01] together with the theorem-level reproduction [ĆMT14, pp. 480–481, Thm. 2.2]. No physical inference is taken from the latter article.

For a self-adjoint boundary operator B, the APS reduced correction is

$$\xi_\rho(B) = \frac{\eta_\rho(B) + h_\rho(B)}{2}, \qquad h_\rho(B) = \dim\ker B_\rho. \tag{8}$$

The kernel term is part of the definition and of the index correction [APS75b, p. 414, Eq. (3.1) and Thm. 3.3]. Eta-only quantities below are always labeled separately.

4.2 The auxiliary formula

For $r \geq 1$, define the finite virtual character

$$E_r = \sum_{m=0}^{2r-2} \operatorname{sgn}(m - r + 1)[V_m \boxtimes V_{2r-2-m}].$$

Let $\langle \rho, E_r \rangle$ denote the corresponding finite character pairing. In the angular-momentum calculation let C_L, C_R be the Casimirs of the two factors of the type (p, q), with $C(V_n) = n(n+2)/4$. The auxiliary correction is as follows.

Theorem 4.1 (All-rank auxiliary eta and kernel)**.** *For every integer $J \geq 1$, free round spherical quotient S^3/Γ, and flat twist ρ,*

$$\eta_\rho(B_J^{\mathrm{aux}}) = \sum_{r=1}^{J-1} \eta_\rho(A_r) - \sum_{r=1}^{J} \langle \rho, E_r \rangle, \tag{9}$$

$$h_\rho(B_J^{\mathrm{aux}}) = \sum_{r=1}^{J} \dim \mathrm{Hom}_\Gamma(\rho, V_{r-1} \boxtimes V_{r-1}). \tag{10}$$

Therefore

$$\eta_\rho(B_J) = \sum_{r=1}^{J} \eta_\rho(A_r) - \sum_{r=1}^{J} \langle \rho, E_r \rangle, \tag{11}$$

$$h_\rho(B_J) = h_\rho(A_J) + h_\rho(B_J^{\mathrm{aux}}). \tag{12}$$

Proof. The diagonal restriction is $T_J|_{\mathrm{Spin}(3)} = \bigoplus_{r=1}^{J} V_{2r-1}$. On its rth summand, (5) and the angular-momentum commutators give

$$D^2 = 2(C_L + C_R) + \tfrac{3}{4} - C_{S+L+R} = 2(C_L + C_R) + 1 - r^2.$$

To fix the signed spectrum, couple $L + R$ first. In a multiplicity-two space the two intermediate spins are $t = r - 1, r$. The diagonal matrix elements of D are respectively $(C_R - C_L)/r$ and $-(C_R - C_L)/r$. For $r = 1, t = 0$ the first is zero, as $C_L = C_R$ there. Hence the trace is zero, and the square law makes the two eigenvalues opposite. They cancel in eta before taking the quotient.

On the one-copy type $(n+2r, n)$ only $t = r$ is available, giving $D = (C_L - C_R)/r = n + r + 1$; the mirror type has the negative eigenvalue. On the finite one-copy type $(m, 2r-2-m)$ only $t = r-1$ is available, giving

$$D = (C_R - C_L)/r = r - 1 - m, \qquad 0 \le m \le 2r - 2.$$

Its signed character is therefore $-E_r$, not E_r. The only zero type is $V_{r-1} \boxtimes V_{r-1}$. Passing to Γ-invariants after twisting by ρ, and then summing in r, gives the full formula. Removing the $r = J$ TT ladder leaves (9); the TT ladders have no kernel, so all zero types remain in (10). The square law alone would not have fixed the relative sign between the infinite and finite branches. □

4.3 Auxiliary characters and cumulative assembly

The generic odd-Chebyshev and representation-ring algebra used in the cumulative calculation is developed in [Mah26a]. It is algebraic input only; the square completion, reducing split, local density, and auxiliary APS formulas are the boundary results proved here. The cumulative corollary below also uses the rankwise Dirac-to-tensor-curl eta formula of [Mah26b], restricted to the common hypotheses of the two papers.

Every multiplicity-two representation has a paired $(+\lambda, -\lambda)$ spectrum and cancels before quotient projection. The unpaired infinite branches are the tensor-curl ladders; what remains is the finite $-E_r$ contribution and the zero representation counted in (10).

The auxiliary eta does not vanish universally. For the trivial sector at ranks $J = 1, 2, 3, 4$, the exact values include

$$S^3/Q_8: \quad 0,\ -\frac{9}{4},\ -\frac{15}{4},\ -\frac{3}{2}, \qquad S^3/(2T): \quad 0,\ -\frac{59}{36},\ -\frac{55}{12},\ -\frac{43}{6}.$$

Here the groups act by right multiplication. Both the signed spectral assembly and direct class averaging reproduce these values. Thus the auxiliary term carries genuine nonabelian representation data.

The higher-spin operators, sphere spectra, twisted quotient formulas, eta invariant, and APS kernel correction are established components. What was not found in the surveyed primary sources is the combined formula (9)–(10) in this natural round-boundary normalization.

4.4 Bounded cumulative eta and reduced correction

Fix the common domain of the cited rankwise transfer and the boundary assembly above: the same free effective round spherical quotient S^3/Γ, injective Spin(4) lift, orientation, operator and copy-multiplicity normalization, and flat twist. Let A_L, A_R be multiplication by the two lifted defining representations in the common representation-ring convention, let U_m be the Chebyshev polynomial of the second kind, and put

$$C_J(A) = U_{J-1}(A/2)U_J(A/2), \qquad (\boldsymbol{e}_r)_\rho = \langle \rho, E_r \rangle, \qquad \boldsymbol{e}_{\le J} = \sum_{r=1}^{J} \boldsymbol{e}_r.$$

Write $\boldsymbol{\eta}_J^B = (\eta_\rho(B_J))_\rho$ and let $\boldsymbol{\eta}_C^D$ be the twisted spin–Dirac eta vector in the convention of [Mah26b]: its positive branch is $V_n \boxtimes V_{n+1}$. This is the opposite of the untwisted Dirac sign

induced by the signed-angular boundary convention of the present paper. The subscript C records the companion convention; the factor $-1/2$ below uses that sign and must not be combined with its negative.

Corollary 4.2 (Cumulative boundary eta and reduced correction)**.** *For every integer $J \geq 1$ on the common domain above,*

$$\boldsymbol{\eta}_J^B = -\frac{1}{2}[C_J(A_L) + C_J(A_R)]\boldsymbol{\eta}_C^D - \boldsymbol{e}_{\leq J}, \tag{13}$$

$$\xi_\rho(B_J) = \frac{1}{2}\left\{\eta_\rho(B_J) + h_\rho(A_J) + \sum_{r=1}^{J} \dim \mathrm{Hom}_\Gamma(\rho, V_{r-1} \boxtimes V_{r-1})\right\}. \tag{14}$$

Proof. The rankwise transfer theorem of [Mah26b] gives $\boldsymbol{\eta}_r^{\mathrm{TT}} = -\frac{1}{2}(A_{2r-1}^L + A_{2r-1}^R)\boldsymbol{\eta}_C^D$. In the common representation-ring convention, $A_m^X = U_m(A_X/2)$. Substitute this formula for each $\eta_\rho(A_r)$ in (11), then apply the two separate Chebyshev identities from [Mah26a], $\sum_{r=1}^{J} U_{2r-1}(A_X/2) = C_J(A_X)$ for $X = L, R$. This gives (13). Equation (14) follows from (8), (12), and (10). □

This corollary is a composition of two cited inputs with the boundary assembly proved here; it is not a new operator intertwining or isospectrality statement. Its hypotheses are exactly the common round quotient, lift, orientation, normalization, and flat-twist hypotheses stated above.

5 Exact round-lens calculations

5.1 Domain and two routes

We next calculate a finite-domain reference table for round lens spaces and ranks. The eta, kernel, and reduced correction values are independently reconstructed by assembly and cyclotomic Fourier routes. The calculation is not an all-level classification or a numerical index on a chosen filling.

The calculation domain is

$$2 \leq p \leq 24, \quad q \in (\mathbb{Z}/p\mathbb{Z})^\times, \quad J = 1, \ldots, 6,$$

with both spin lifts where distinct and every flat character $a \in \mathbb{Z}/p\mathbb{Z}$. It comprises 242 lens/lift cells, 1,452 rank rows, and 23,688 exact triples

$$(\eta_a(B_J), h_a(B_J), \xi_a(B_J)).$$

Route A uses the all-rank split (9)–(12), the exact Hurwitz/quasipolynomial tensor-curl formula, and direct weight enumeration for the finite terms. Route B starts from the closed geometric-twist class transform in Section 6, recovers per-rank blocks by transform differences, and applies exact finite Fourier inversion. The implementation works in $\mathbb{Q}[X]/(1 + X + \cdots + X^{p-1})$, which retains all nonidentity pth roots, including imprimitive ones; the odd lift makes both character exponents integral. Route B uses no Hurwitz continuation. At the identity class the eta coefficient is zero by left-right spectral symmetry, while kernel and finite terms are exact dimension counts.

Across the finite domain the routes agree in 94,752 exact value checks. The complete tables are supplied as ancillary machine-readable data, with both route implementations and exact assertion-driven checks. Two further out-of-domain cells are supplied as additional reproducibility checks, not as unobserved predictions.

5.2 Selected values and independent checks

Table 1 records a small human-readable selection. The $L(17,3)$ row checks the auxiliary formula directly; the other rows were calculated outside the working domain without changing the formulas.

space	J	a	$\eta_a(B_J)$	$h_a(B_J)$	$\xi_a(B_J)$
$L(17,3)$	6	5	$-9/17$	6	93/34
$L(17,3)$	6	12	$-9/17$	6	93/34
$L(29,12)$	7	3	$-13/29$	5	66/29
$L(29,12)$	7	26	$-13/29$	5	66/29
$L(27,7)$	8	9	$-1/9$	7	31/9

Table 1: Selected exact values from the assembly and Fourier calculations.

At $L(17,3)$, $J=6$, $a=5$ and its conjugate $a=12$, the auxiliary values are $\eta(B_6^{\mathrm{aux}})=-8/17$ and $h(B_6^{\mathrm{aux}})=6$, while the full eta is $-9/17$. Conjugation, orientation reversal, spin-lift invariance, exact denominator typing, and the identities

$$\sum_a \eta_a(B_J)=0, \qquad \sum_a h_a(B_J)=\frac{J(J+1)(2J+1)}{6}$$

are verified throughout the table domain.

6 Class transforms and oriented reconstruction

Let $L(p,q)$ carry the fixed odd lift $\widetilde{q}$. For a nonidentity cyclic class represented by k, $1\le k\le p-1$, put

$$\zeta=e^{\pi i/p}, \qquad e_L=k(1+\widetilde{q}), \qquad e_R=k(1-\widetilde{q}),$$
$$P_k=(\zeta^k-\zeta^{-k})(\zeta^{k\widetilde{q}}-\zeta^{-k\widetilde{q}}).$$

For the $SU(2)$ character

$$\chi_m(z)=\sum_{j=0}^{m} z^{m-2j}=\frac{z^{m+1}-z^{-(m+1)}}{z-z^{-1}},$$

with the polynomial value used also at $z=\pm1$, write $\chi_m^L=\chi_m(\zeta^{e_L})$ and $\chi_m^R=\chi_m(\zeta^{e_R})$. On nonidentity free classes, exact Fourier inversion of the complete flat-twist vectors gives

$$\widehat{\eta}_J^{\mathrm{full}}(k)=\frac{\chi_J^L\chi_{J-1}^R+\chi_{J-1}^L\chi_J^R}{P_k}. \tag{15}$$

To see the relative sign algebraically, put $x=\chi_1^L$, $y=\chi_1^R$, so $P_k=x-y$. The recurrence $\chi_{m+1}(x)=x\chi_m(x)-\chi_{m-1}(x)$ telescopes the finite character to

$$P_k\sum_{r=1}^{J}E_r=(\chi_J^L-\chi_J^R)(\chi_{J-1}^L-\chi_{J-1}^R).$$

The positive-TT cumulative numerator is $\chi_J^L\chi_{J-1}^L+\chi_J^R\chi_{J-1}^R$. Subtracting the finite term gives exactly the cross numerator in (15).

The kernel character is

$$H_J = \sum_{r=0}^{J-1} \chi_r^L \chi_r^R.$$

Christoffel–Darboux gives

$$P_k H_J = \chi_J^L \chi_{J-1}^R - \chi_{J-1}^L \chi_J^R.$$

Combining eta and kernel according to (8) selects one chirality:

$$\widehat{\xi}_J(k) = \frac{\chi_J^L \chi_{J-1}^R}{P_k}. \tag{16}$$

6.1 Blind classes

The full-eta blind set is the cyclotomic divisibility locus of

$$\chi_J^L \chi_{J-1}^R + \chi_{J-1}^L \chi_J^R.$$

For every even p, the half-turn class $k = p/2$ is blind for every rank and lift. Thus no finite collection of full-eta ranks has complete Fourier support at all levels.

The reduced transform vanishes precisely when

$$(J+1)e_L \equiv 0 \pmod{p}, \qquad e_L \not\equiv 0, p \pmod{2p},$$

or

$$Je_R \equiv 0 \pmod{p}, \qquad e_R \not\equiv 0, p \pmod{2p}.$$

Reduced ranks one and two have no common blind nonidentity class for any p, q, or lift. This is a universal support statement, not by itself a reconstruction theorem.

6.2 The known rank-one base

At $J = 1$, the auxiliary eta is zero and the kernel vector is $h_1 = (1, 0, \ldots, 0)$ in flat-character coordinates. Hence

$$\eta_1 = 2\xi_1 - h_1$$

is a q-independent invertible translation. The rank-one twist-eta reconstruction theorem is known: under one coherently typed multiplicative character relabeling, equality of the full twist vector forces $q' \equiv q^{\pm 1} \pmod{p}$ [Kat90, pp. 23–24, Thm. 2.1; pp. 34–35, Thm. 4.3].

6.3 Adjacent-rank reconstruction is refuted

Do two adjacent ranks, each retaining the full coherently labelled twist vector, determine the oriented lens space? The answer is no. The all-rank multipliers, chiral reduction, and support algebra above do not promote to universal oriented reconstruction from two adjacent ranks. Under our convention, oriented equivalence of lens spaces requires $q' \equiv q^{\pm 1} \pmod{p}$, while unoriented homeomorphism allows $q' \equiv \pm q^{\pm 1} \pmod{p}$; this is the classical topological classification perimeter [Bro60].

The adjacent-rank calculation yields 16 witness rows, all recomputed by both exact routes in the present convention. They represent four pairs of colliding oriented orbits:

p	ranks	left orbit	right orbit	rows
7	$(6,7)$	$\{2,4\}$	$\{3,5\}$	4
14	$(6,7)$	$\{3,5\}$	$\{9,11\}$	4
27	$(8,9)$	$\{4,7\}$	$\{16,22\}$	4
27	$(8,9)$	$\{5,11\}$	$\{20,23\}$	4

Every row refutes oriented reconstruction. The rows at $p = 7$ and $p = 14$ belong to the blind-rank families proved in Section 7; their rank vectors carry no information about q. Those eight rows also merge under orientation reversal, so they are orientation-reversing homeomorphic. The eight rows at $p = 27$, representing two orbit-pair collisions, remain nonhomeomorphic even unoriented. In particular, the adjacent-rank vectors for $L(27, 4)$ and $L(27, 22)$ agree exactly with no character relabeling.

A representative of the second $p = 27$ pair is $L(27, 5)$ versus $L(27, 20)$ with the same relabeling applied to both the reduced vectors and the kernels: $a \mapsto 7a$ (also $a \mapsto 20a$).

On the computed domain, the divisibility/size pattern $(J + 1) \mid p$ together with $p \le 3(J + 1)$ is an observation, not a theorem. A refined positive reconstruction criterion remains open. This negative result is also distinct from the phase collapse below: $L(16, 1)$ and $L(16, 9)$ are exactly distinguished at every computed adjacent pair $(1, 2), \dots, (8, 9)$. Their all-rank phase loss is therefore not exact-value loss on this range.

7 Exponentiated reduced correction and information loss

Exponentiation retains the class of the reduced correction modulo integers:

$$\tau(B) = \exp(2\pi i\, \xi(B)).$$

The reduced eta and determinant-line formalism, including the deliberate loss of the integer lift, is classical [DF94, arXiv v4, pp. 1, 6–8, 12–13]. Fractional eta-type information is related to torsion linking pairings [SS02, arXiv v2, pp. 2, 15, 18]. Ordinary Dirac spectra can also lose geometric information on spherical space forms [Bär96, pp. 80–82, Thm. 5]. Current second-jet equivariant eta separations retain different data and concern different lens pairs [Sha26, Thm. 1.1, Thm. 4.2, Thm. 5.14, Cor. 5.30].

7.1 Transform assumptions and Fourier–Dedekind reduction

Throughout this section, $p \ge 2$, q is a unit modulo p, the lift $\widetilde{q}$ is odd, $u = q^{-1} \pmod p$, $N = J(J+1)$, and $J \ge 1$. We take (16) on nonidentity classes and $\widehat{\xi}_J(0) = N(2J+1)/12$ as the defining data. No filling or unresolved geometric sign choice enters the following proofs. Set $\alpha = (1 + \widetilde{q})/2$, $\beta = (1 - \widetilde{q})/2$ and

$$G(x) = x^{\alpha} \chi_J(x^{\alpha}) \chi_{J-1}(x^{\beta}) = \sum_e g_e x^e.$$

The factorization $P_k = x^{-\alpha}(1 - x)(1 - x^q)$ at $x = \exp(2\pi i k/p)$ gives the exact Fourier inversion

$$\xi_{J,a} = \frac{N(2J+1)}{12p} + \sum_e g_e S_{e+a}, \qquad S_m = \frac{1}{p} \sum_{\substack{\lambda^p = 1 \\ \lambda \ne 1}} \frac{\lambda^m}{(1 - \lambda)(1 - \lambda^q)}. \tag{17}$$

Fourier–Dedekind sums are classical; see [BR04]. Define the ordinary Dedekind sum by $s(q,p) = \sum_{j=1}^{p-1}(j/p - 1/2)((qj \bmod p)/p - 1/2)$, where remainders are least nonnegative integers. Direct finite summation gives

$$S_m = \frac{1}{p^2}\sum_{j=0}^{p-1} j\,((-m-qj) \bmod p) - \frac{(p-1)^2}{4p}, \tag{18}$$

$$S_0 = -s(q,p) + \frac{p-1}{4p}, \qquad S_{m+1} - S_m = \frac{(-mu \bmod p)}{p} - \frac{p-1}{2p}. \tag{19}$$

For example, the first identity follows from $(\lambda-1)\sum_{j=0}^{p-1} j\lambda^j = p$ at nonidentity roots and orthogonality of the pth roots. Summing the difference, also backwards for negative m, yields

$$S_m - S_0 \equiv Q(m) := -\frac{um(m-1) + (p-1)m}{2p} \pmod{\mathbb{Z}}. \tag{20}$$

7.2 The phase law and rank-one generation

Theorem 7.1 (All-rank phase formula)**.** *For the stated transform, every rank and flat label satisfy*

$$\xi_{J,a} - \xi_{J,0} \equiv -\frac{Nua^2}{2p} \pmod{\mathbb{Z}}, \tag{21}$$

$$\xi_{J,a} \equiv -Ns(q,p) - \frac{Nua^2}{2p} - \frac{N(N-2)(\widetilde{q}+u)}{12p} - \frac{N}{4} \pmod{\mathbb{Z}}. \tag{22}$$

Both spin lifts give the same phases.

Proof. The first three moments of G are

$$G_0 := \sum_e g_e = N, \qquad G_1 := \sum_e e g_e = \alpha N,$$

$$G_2 := \sum_e e^2 g_e = \alpha^2 N + \frac{\alpha^2 J^2(J+1)(J+2) + \beta^2 J(J+1)^2(J-1)}{3}.$$

The character's first logarithmic derivative at one vanishes; its second is $\gamma^2 r(r+1)(r+2)/3$ for $\chi_r(x^\gamma)$. Equations (17)–(20) give the relative phase as $-Nua^2/(2p)$ plus $-aN(u\widetilde{q} - 1 + p)/(2p)$, which is integral. For the absolute phase, the same equations give

$$\xi_{J,0} \equiv \frac{N(2J+1)}{12p} + NS_0 - \frac{u(G_2 - G_1) + (p-1)G_1}{2p} \pmod{\mathbb{Z}}.$$

Put $t = \widetilde{q}$ and $ut = 1 + pk$. Substitution reduces the difference between this expression and the right side of (22) at $a = 0$ to integers: the nontrivial term is $Nk[t(N+1) + 2J + 1]/12$. Indeed $12 \mid N[t(N+1) + 2J + 1]$: modulo three either $3 \mid N$ or $J \equiv 1$ and the bracket is zero; modulo four either $4 \mid N$ or $N \equiv 2$ and the bracket is even. Also $N(t-1)/4$ is integral. Finally $24 \mid N(N-2) = (J-1)J(J+1)(J+2)$, so changes of lift or of the representative of u change the formula by integers. □

The classical congruence $12p\,s(q,p) \equiv q + u \pmod{p}$, including the integrality of $12p\,s$, is an arithmetic property of Dedekind sums [RG72, Ch. 3]. It simplifies (22) to

$$\xi_{J,a} \equiv -N(N-1)s(q,p) - \frac{Nua^2}{2p} - \frac{N}{4} \pmod{\mathbb{Z}}. \tag{23}$$

Corollary 7.2 (Rank one generates all phases)**.** *With $c_J = J(J+1)/2$ and $\tau_{J,a} = \exp(2\pi i \xi_{J,a})$,*

$$\xi_{J,a} \equiv c_J \xi_{1,a} + 2c_J(c_J - 1)\xi_{1,0} \pmod{\mathbb{Z}}, \tag{24}$$

$$\tau_{J,a} = \tau_{1,a}^{c_J} \tau_{1,0}^{2c_J(c_J-1)}. \tag{25}$$

Thus the complete exponentiated rank ladder carries no more information than the labelled rank-one phases, under the same coherent relabeling.

Proof. Insert $\xi_{1,a} \equiv -2s - ua^2/p - 1/2$ in (23). The remaining constant difference is $N(N-2)/4 \in \mathbb{Z}$. □

This does not identify the full real-valued corrections, nor eliminate the information in their integer lifts.

7.3 A complete phase-collapse criterion

Theorem 7.3 (Coherent-label collapse)**.** *For the same modulus p, units q, q', n, and any allowed lifts, $\xi_{J,a}(p,q) \equiv \xi_{J,na}(p,q') \pmod{\mathbb{Z}}$ for every J, a if and only if*

$$q' \equiv n^2 q \pmod{p}, \qquad 2s(q,p) \equiv 2s(q',p) \pmod{\mathbb{Z}}.$$

Proof. At rank one, label zero imposes the Dedekind condition. Subtracting that case from label one imposes $u \equiv n^2 u' \pmod{p}$, equivalent to the first condition. Conversely these conditions give equality at every label of rank one; Corollary 7.2 gives every rank. □

The first condition is the usual oriented homotopy relation, weaker than oriented homeomorphism. The second is additional arithmetic information. Equality questions modulo integers for Dedekind sums have an established literature [Tsu14]; the theorem is an application to the specified transform, not a new theory of Dedekind sums.

Corollary 7.4 (The $p = 16$ example)**.** *The pair $L(16,1)$, $L(16,9)$ is not orientation-preservingly homeomorphic, but has identical phase ladders under $a \mapsto 5a$ (also $3a, 11a, 13a$).*

Proof. The oriented homeomorphism criterion fails since $9 \neq 1^{\pm 1}$ modulo 16. But $5^2 \equiv 9$ and $12 \cdot 16\, s(1,16) = 210$, $12 \cdot 16\, s(9,16) = 18$; their difference is 192, divisible by $6p$. Apply Theorem 7.3. □

The separate quadratic-in-rank calculation and its 1,536 integer coefficients remain in the supplement as an independent finite certificate of this example, not the proof of the general theorem.

7.4 Infinite blind adjacent-rank families

The following statements concern the exact ξ values, not just phases.

Theorem 7.5 (Blind families, including composite moduli)**.** *Let $m \geq 5$ be odd and $\gcd(q,m) = \gcd(q^2-1, m) = 1$. For $p = m$ and $r \in \{m-1, m\}$,*

$$\xi_{r,a} = \frac{r(r+1)(2r+1)}{12p} \qquad \textit{for all } a.$$

For $p = 2m$, take q odd with the same conditions modulo m. Then

$$\xi_{r,a} = \frac{1}{p}\left\{\frac{r(r+1)(2r+1)}{12} + (-1)^{r-1+a}\frac{r(r+1)}{4}\right\}, \quad r = m-1, m.$$

In either family the two rank vectors are independent of the allowed q.

Proof. Writing $z = \zeta^{e_L}$ and $w = \zeta^{e_R}$, centrality means $p \mid k(1+\widetilde{q})$ or $p \mid k(1-\widetilde{q})$. The gcd hypotheses leave only $k = 0$ for $p = m$, and $k = 0, m$ for $p = 2m$. Since $p \mid 2m$, $\chi_{m-1}(z) = 0$ whenever $z \neq \pm 1$, and likewise for w. Thus the reduced transforms at both ranks vanish on all noncentral classes. For $p = m$ only the identity term remains. For $p = 2m$, at $k = m$ one has $w = -z$; substitution in the character polynomials gives $\widehat{\xi}_r(m) = (-1)^{r-1}r(r+1)/4$. Fourier inversion multiplies this by $(-1)^a/p$, as asserted. □

For odd $m \geq 5$, an admissible q exists exactly when $3 \nmid m$. If $3 \mid m$, every unit has $q^2 \equiv 1 \pmod 3$, contradicting the hypothesis. Conversely, $q = 2$ works when $3 \nmid m$; for the $p = 2m$ family, its odd lift $q = m + 2$ works. Primality is not necessary: the hypotheses hold at $m = 25, 35, 49$ with $q = 2$, and with $q = 3$ at $m = 25, 35$. The prime subfamily includes infinitely many nonhomeomorphic collisions: for primes $p \geq 11$, the $p - 3$ allowed residues cannot all lie in one unoriented homeomorphism orbit, whose size is at most four. The partial-blindness cases $p = 3(J+1)$ lie outside these fully blind families; their classification and a general positive exact-value reconstruction criterion are not settled by this theorem.

Finite controls, not additional theorem hypotheses. The phase formula and rank-one identity each reproduce all 23,688 supplied table values. The quotient-ring route also checks 3,644 selected prime and twice-prime entries and 1,690 composite-family entries. These are reproduced controls, not prospective predictions. Representative-pair censuses distinguish orientation-preserving from unoriented equivalence; they must not be read as counts after quotienting every homeomorphism orbit.

Selected exact lifts illustrate the equality modulo integers.

J	a	$5a \bmod 16$	$\xi_{J,a}(16,1)$	$\xi_{J,5a}(16,9)$	difference
1	1	5	$-7/4$	$1/4$	-2
2	1	5	$-7/2$	$1/2$	-4
6	1	5	$-1/4$	$11/4$	-3

Table 2: Exact phase-lift values for one flat label. Integer differences give identical exponentials.

The kernel term is essential. The eta-only comparison $[\eta/2]$ separates the same pair already at rank two, while the genuine APS class $[(\eta+h)/2]$ does not. Hence the real eta vector, eta-only reduction, and exponentiated APS correction retain genuinely different information. The collapse criterion classifies equality of this exponentiated transform under coherent relabeling, not homeomorphism of lens spaces. It does not classify equality of the full real-valued invariants.

8 Conclusion and open problems

The full Dirac completion and its sparse predecessor must be distinguished, even though their interior symbols are elliptically homotopic. For the full completion the first-order homogeneous derivation fixes the signed spectrum and hence the character transform. The auxiliary sector cannot be discarded: it contributes both spectral asymmetry and zero modes, and the latter can change what survives after exponentiation. At the level of real class transforms the data can be strongly reconstructive; at the level of exponentiated reduced corrections, the pair $L(16, 1)$ and $L(16, 9)$ shows an all-rank loss of oriented information. More strongly, the entire phase ladder is generated by rank one, whereas selected adjacent real-valued ranks can be completely blind to the lens parameter. These are distinct information losses.

Several natural questions remain. The fixed-point formula must use the actual source and target fiber characters [Gil84b, Sec. 4.5]; the character check in Section 3.1 is complementary to, not a replacement for, the first-order operator derivation. The present calculation does not choose a four-dimensional filling and therefore gives no numerical APS index. A positive reconstruction criterion beyond the rank-one theorem is still open, despite the adjacent-rank counterexamples found here. It is also unknown how much of the reducing split and phase law persists for Berger or more general off-round metrics. Finally, determinant magnitudes require information not contained in the reduced correction. These are analytic and geometric questions; no physical interpretation is attached to the phase notation used in this paper.

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